



## SAFETY DATA SHEET

### RETAIL POWER XL

#### SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : RETAIL POWER XL

Other means of identification : Not applicable

Recommended use : Machine Warewashing Detergent

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.4 %

Company : Kay Chemical Company  
8300 Capital Drive  
Greensboro, North Carolina USA 27409  
1-800-529-5458

Emergency telephone : 1-800-328-0026 (US/Canada), 1-651-222-5352 (outside US)

Issuing date : 04.06.2015

#### SECTION 2. HAZARDS IDENTIFICATION

##### Product AS SOLD

Physical and chemical hazards : Not flammable or combustible.

##### Potential Health Effects

Eyes : Causes eye irritation.

Skin : Causes mild skin irritation.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

##### Product AT USE DILUTION

Physical and chemical hazards : Not flammable or combustible.

##### Potential Health Effects

Eyes : Causes eye irritation.

Skin : Causes mild skin irritation.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

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#### SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

##### Product AS SOLD

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Pure substance/mixture : Mixture

| Chemical Name                          | CAS-No.    | Concentration (%) |
|----------------------------------------|------------|-------------------|
| sodium hydroxide                       | 1310-73-2  | 60 - 100          |
| Sodium Carbonate(soda)                 | 497-19-8   | 5 - 10            |
| oxirane, methyl-, polymer with oxirane | 9003-11-6  | 1 - 5             |
| Polycarboxylates                       | 52255-49-9 | 1 - 5             |

#### Product AT USE DILUTION

No hazardous ingredients

### SECTION 4. FIRST AID MEASURES

#### Product AS SOLD

|                                                             |                                                                                |
|-------------------------------------------------------------|--------------------------------------------------------------------------------|
| In case of eye contact                                      | : Rinse with plenty of water.                                                  |
| In case of skin contact                                     | : Rinse with plenty of water.                                                  |
| If swallowed                                                | : Rinse mouth. Get medical attention if symptoms occur.                        |
| If inhaled                                                  | : Get medical attention if symptoms occur.                                     |
| Protection of first-aiders                                  | : No special precautions are necessary for first aid responders.               |
| Notes to physician                                          | : Treat symptomatically.                                                       |
| Most important symptoms and effects, both acute and delayed | : See Section 11 for more detailed information on health effects and symptoms. |

#### Product AT USE DILUTION

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| In case of eye contact  | : Rinse with plenty of water.                           |
| In case of skin contact | : Rinse with plenty of water.                           |
| If swallowed            | : Rinse mouth. Get medical attention if symptoms occur. |
| If inhaled              | : Get medical attention if symptoms occur.              |

### SECTION 5. FIRE-FIGHTING MEASURES

#### Product AS SOLD

|                                       |                                                                                                                                                  |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media          | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                        |
| Unsuitable extinguishing media        | : None known.                                                                                                                                    |
| Specific hazards during fire fighting | : Not flammable or combustible.                                                                                                                  |
| Hazardous combustion products         | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus |

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Special protective equipment for fire-fighters : Use personal protective equipment.

Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

#### SECTION 6. ACCIDENTAL RELEASE MEASURES

##### Product AS SOLD

Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Sweep up and shovel into suitable containers for disposal.

##### Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### SECTION 7. HANDLING AND STORAGE

##### Product AS SOLD

Advice on safe handling : Wash hands thoroughly after handling.

Conditions for safe storage : Do not store near acids. Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : -10 °C to 50 °C

##### Product AT USE DILUTION

Advice on safe handling : Wash hands thoroughly after handling.

Conditions for safe storage : Do not store near acids. Keep out of reach of children. Store in suitable labeled containers.

#### SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

##### Product AS SOLD

##### Ingredients with workplace control parameters

| Ingredients      | CAS-No.   | Form of exposure | Permissible concentration | Basis     |
|------------------|-----------|------------------|---------------------------|-----------|
| sodium hydroxide | 1310-73-2 | C                | 2 mg/m <sup>3</sup>       | ACGIH     |
|                  |           | Ceiling          | 2 mg/m <sup>3</sup>       | NIOSH REL |

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|  |  |     |         |         |
|--|--|-----|---------|---------|
|  |  | TWA | 2 mg/m3 | OSHA Z1 |
|--|--|-----|---------|---------|

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

#### Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling.

#### Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

#### Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

### SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         | Product AS SOLD     |
|-----------------------------------------|---------------------|
| Appearance                              | : Cast Solid        |
| Color                                   | : white             |
| Odor                                    | : odorless          |
| pH                                      | : 12.5 - 12.8, 1 %  |
| Flash point                             | : Not applicable    |
| Odor Threshold                          | : No data available |
| Melting point/freezing point            | : No data available |
| Initial boiling point and boiling range | : No data available |
| Evaporation rate                        | : No data available |
| Flammability (solid, gas)               | : No data available |
| Upper explosion limit                   | : No data available |
| Lower explosion limit                   | : No data available |
| Vapor pressure                          | : No data available |
| Relative vapor density                  | : No data available |

| Product AT USE DILUTION |
|-------------------------|
| liquid                  |
| colorless               |
| odorless                |
| 12.6                    |

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|                                        |                     |
|----------------------------------------|---------------------|
| Relative density                       | : No data available |
| Water solubility                       | : No data available |
| Solubility in other solvents           | : No data available |
| Partition coefficient: n-octanol/water | : No data available |
| Autoignition temperature               | : No data available |
| Thermal decomposition                  | : No data available |
| Viscosity, kinematic                   | : No data available |
| Explosive properties                   | : No data available |
| Oxidizing properties                   | : No data available |
| Molecular weight                       | : No data available |
| VOC                                    | : No data available |

### SECTION 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                                    |                                                                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical stability                 | : Stable under normal conditions.                                                                                                                |
| Possibility of hazardous reactions | : No dangerous reaction known under conditions of normal use.                                                                                    |
| Conditions to avoid                | : None known.                                                                                                                                    |
| Incompatible materials             | : Acids<br>Metals<br>Organic materials                                                                                                           |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides<br>Nitrogen oxides (NOx)<br>Sulfur oxides<br>Oxides of phosphorus |

### SECTION 11. TOXICOLOGICAL INFORMATION

|                                          |                             |
|------------------------------------------|-----------------------------|
| Information on likely routes of exposure | : Eye contact, Skin contact |
|------------------------------------------|-----------------------------|

#### Potential Health Effects

#### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes eye irritation.                                      |
| Skin             | : Causes mild skin irritation.                                |
| Ingestion        | : Health injuries are not known or expected under normal use. |
| Inhalation       | : Health injuries are not known or expected under normal use. |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Product AT USE DILUTION

|      |                          |
|------|--------------------------|
| Eyes | : Causes eye irritation. |
|------|--------------------------|

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|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Skin             | : Causes mild skin irritation.                                |
| Ingestion        | : Health injuries are not known or expected under normal use. |
| Inhalation       | : Health injuries are not known or expected under normal use. |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Experience with human exposure

##### Product AS SOLD

|              |                                  |
|--------------|----------------------------------|
| Eye contact  | : Redness, Irritation            |
| Skin contact | : slight irritation              |
| Ingestion    | : No symptoms known or expected. |
| Inhalation   | : No symptoms known or expected. |

##### Product AT USE DILUTION

|              |                                  |
|--------------|----------------------------------|
| Eye contact  | : Redness, Irritation            |
| Skin contact | : slight irritation              |
| Ingestion    | : No symptoms known or expected. |
| Inhalation   | : No symptoms known or expected. |

#### Toxicity

##### Product AS SOLD

|                                   |                        |
|-----------------------------------|------------------------|
| Acute oral toxicity               | : No data available    |
| Acute inhalation toxicity         | : No data available    |
| Acute dermal toxicity             | : No data available    |
| Skin corrosion/irritation         | : Mild skin irritation |
| Serious eye damage/eye irritation | : Mild eye irritation  |
| Respiratory or skin sensitization | : No data available    |
| Carcinogenicity                   | : No data available    |
| Reproductive effects              | : No data available    |
| Germ cell mutagenicity            | : No data available    |
| Teratogenicity                    | : No data available    |
| STOT-single exposure              | : No data available    |
| STOT-repeated exposure            | : No data available    |
| Aspiration toxicity               | : No data available    |

##### Ingredients

|                     |                                                   |
|---------------------|---------------------------------------------------|
| Acute oral toxicity | : Sodium Carbonate(soda)<br>LD50 Rat: 2,800 mg/kg |
|                     | Polycarboxylates<br>LD50 Rat: > 5,000 mg/kg       |

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#### Ingredients

Acute dermal toxicity : Polycarboxylates  
LD50 Rabbit: > 5,000 mg/kg

### SECTION 12. ECOLOGICAL INFORMATION

#### Product AS SOLD Ecotoxicity

Environmental Effects : Harmful to aquatic life.

#### Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

#### Ingredients

Toxicity to fish : Sodium Carbonate(soda)  
96 h LC50 *Lepomis macrochirus* (Bluegill sunfish): 300 mg/l

oxirane, methyl-, polymer with oxirane  
96 h LC50 Fish: > 100 mg/l

Polycarboxylates  
96 h LC50 Fish: > 100 mg/l

#### Ingredients

Toxicity to daphnia and other aquatic invertebrates : sodium hydroxide  
48 h EC50: 40 mg/l

Sodium Carbonate(soda)  
48 h EC50 *Ceriodaphnia* (water flea): 213.5 mg/l

#### Persistence and degradability

No data available

#### Bioaccumulative potential

No data available

#### Mobility in soil

No data available

#### Other adverse effects

No data available

### SECTION 13. DISPOSAL CONSIDERATIONS

#### Product AS SOLD

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

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Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

#### Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

### SECTION 14. TRANSPORT INFORMATION

#### Sea transport (IMDG/IMO)

UN number : 1823  
Description of the goods : SODIUM HYDROXIDE, SOLID  
Class : 8  
Packing group : II  
Marine pollutant : no

#### Product AT USE DILUTION

Not intended for transport.

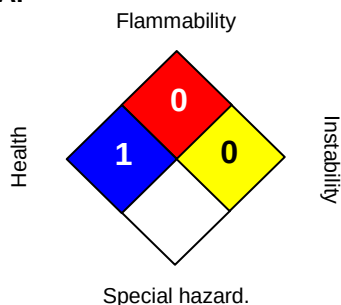
### SECTION 15. REGULATORY INFORMATION

For Regulatory information please contact your Ecolab sales representative.

### SECTION 16. OTHER INFORMATION

#### Product AS SOLD

##### NFPA:



##### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 1 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

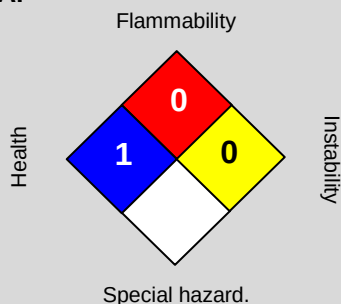
#### Product AT USE DILUTION



## SAFETY DATA SHEET

### RETAIL POWER XL

#### NFPA:



#### HMIS III:

|                 |   |
|-----------------|---|
| HEALTH          | 1 |
| FLAMMABILITY    | 0 |
| PHYSICAL HAZARD | 0 |

0 = not significant, 1 = Slight,  
2 = Moderate, 3 = High  
4 = Extreme, \* = Chronic

Issuing date : 04.06.2015  
Version : 1.2  
Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

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